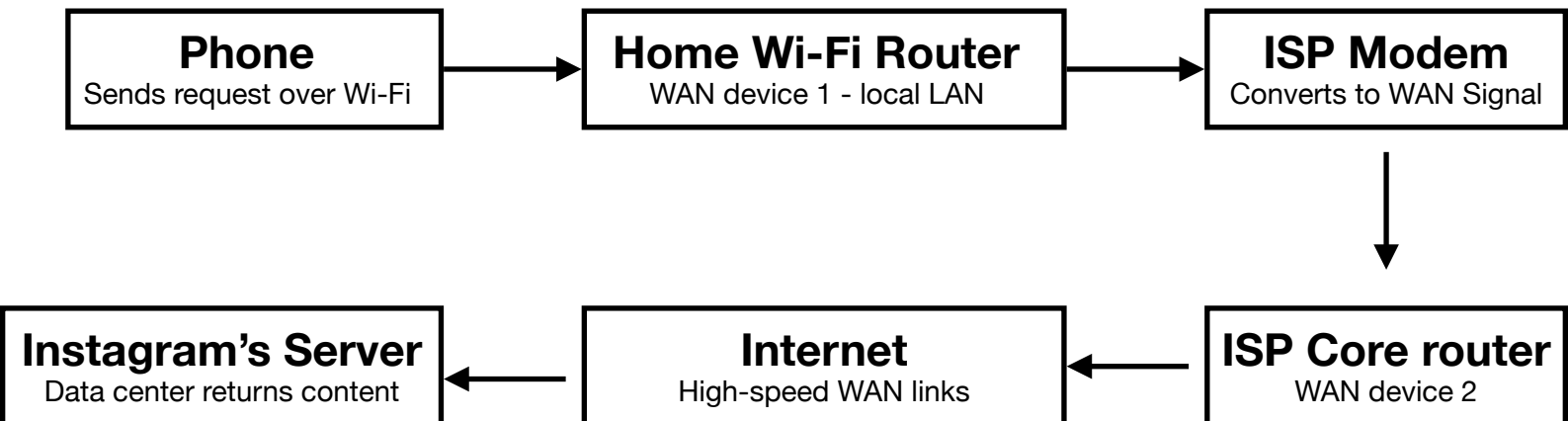


Module 2: Network + Network Security , Maintenance and Troubleshooting

Wide Area Network

Task 1:



Task 2:

WAN Type	Speed	Cost	Use Case
MPLS (Multi-Protocol Label Switching)	High, consistent, low latency	Expensive (per-site, per-Mbps pricing)	Large enterprises needing guaranteed performance between offices — e.g., a bank connecting branches, or a company like Zomato linking its regional data centers

WAN Type	Speed	Cost	Use Case
Leased Line	Very high, dedicated bandwidth	Very expensive	Mission-critical, always-on links — e.g., IRCTC's ticketing servers connecting to railway zone offices, where downtime isn't acceptable
Broadband Internet	Moderate to high, but shared and variable	Cheap, widely available	Home and small business use — e.g., your home WiFi, a small local kirana store's card-swipe machine connection

Task 3:

When you connect to coffee-shop WiFi, your device joins that shop's local network, which then reaches the internet (the WAN) through their router and ISP — a path you don't control or trust. A VPN app on your phone or laptop encrypts your data *before* it leaves your device, then sends it through the coffee shop's WAN connection to a VPN server elsewhere (often in another city or country).

This creates what's called a "tunnel" — imagine a sealed pipe running through the open public network. Even though your encrypted data physically travels over the same shared WAN infrastructure as everyone else's traffic, anyone intercepting it (like someone snooping on the same WiFi) only sees scrambled data, not your actual Spotify login or YouTube activity. The VPN server then decrypts your request and forwards it to Spotify or YouTube on your behalf, making it look like the request came from the VPN server's location, not the coffee shop.

So the VPN doesn't change the physical WAN path — it changes what travels through it, wrapping your data in an encrypted "envelope" for that leg of the journey.

Task 4:

MPLS

Predictable performance for time-sensitive operations — Flipkart's warehouses need to sync inventory, order, and dispatch data in near real-time between Ahmedabad, Mumbai, and Delhi. MPLS offers low, consistent latency and can prioritize critical traffic (like order-processing data) over less urgent traffic, which plain broadband can't guarantee since it's shared and best-effort.

Reliability and security for business-critical data — Warehouse operations (stock levels, dispatch scheduling) can't tolerate downtime or interception. MPLS provides a private, dedicated-feeling network (even though it runs over shared carrier infrastructure) with built-in redundancy and failover paths, making it more dependable than broadband and considerably cheaper to scale across three cities than individual leased lines everywhere.